Activation Block – Comparison Document

1. Introduction

The objective of this document is to define the architecture for the 8-bit Activation Block within the PE. This block requires a memory-based Look-Up Table (LUT) implementation and synthesizable logic to support complex non-linear functions. We have defined three potential architectural topologies for evaluation: SRAM, Unified ROM, Split (Banked) ROM; see Section 2 for details.

The final architectural selection will be determined by a comparative analysis of Power, Performance, and Area (PPA) metrics, which will be obtained using Synopsys Fusion Compiler.

2. Architecture Options

This section details the three architectural topologies evaluated for the Activation Block. In all configurations, the block is designed to support 6 distinct activation functions.

2.1. **Option A: SRAM (Flexible RAM)**

This option utilizes a single SRAM block (256 rows × 8 bits). The RAM serves as a fully programmable LUT. This architecture offers maximum flexibility. Unlike the ROM options, the content is not fixed at fabrication; the user can load any arbitrary activation function into the memory, allowing for post-silicon updates or support for custom functions.

2.2. **Option B: ROM – Unified Memory**

This option utilizes a single, monolithic ROM block (1024 rows × 8 bits) to store the four non-linear functions: Tanh, Sigmoid, GELU, SiLU (see Table 2). To optimize memory usage, ReLU and Identity are not stored in the ROM. Instead, they are implemented via a parallel synthesizable logic path. The specific function is accessed by concatenating the activation selection bits with the input data to form the address in the unified block.

2.3. **Option C: ROM – Split (Banked) Memory**

This option physically divides the memory into four independent banks (4 banks of 256 rows × 8 bits), each dedicated to one of four non-linear functions: Tanh, Sigmoid, GELU, SiLU (see Table 2). Like the Unified option, ReLU and Identity are offloaded to synthesizable logic.

To minimize area overhead, a global decoder is shared across all four banks.

The architecture utilizes the activation selection signal as an enable mechanism; Only the specific ROM bank corresponding to the chosen activation is enabled.

Because only the relevant bank is active, only the bitlines of that specific bank are charged and driven.

3. Memory Architecture Comparison

3.1. Below is a comparison table of relevant metrics.

Memory Type/ Feature	SRAM	ROM: Unified Memory	ROM: Split (Banked) Memory
Configuration	Single Block (256 rows × 8 bits)	Single Block (1024 rows × 8 bits)	4 Small Banks (4 × [256 rows × 8 bits])
Est. Transistors Count	≈ 17,600	≈ 29,000	≈ 14,000
Transistors per Bit	≈ 8.6 (expensive)	≈ 3.5 (moderate)	≈ 1.7 (highly efficient)
Decoder	Standard 8-to-256	Complex 10-to-1024	Shared 8-to-256
Dynamic Power *(Read Only)*	Moderate (Requires differential precharge of two bitlines per bit)	Medium-High (High parasitic capacitance due to long 1024-cell bitlines)	Lowest (Low capacitance; single-ended sensing of short bitlines)
Static Power	High (leakage from 256 rows)	Low	Low
Delay	Very Fast (uses sense amplifier)	Slow (high bitline capacitance)	Fast (low bitline capacitance)

Table 1: Memory Type/Feature Comparison.

3.2. **Transistor Count Analysis**

The estimated transistor counts presented in Table 1 represent the aggregate of the storage core (bitcells) and the peripheral circuitry (overhead). The following breakdown details the methodology used to derive these estimates for each architecture.

1. SRAM (Single Block: 256 rows × 8 bits)

- Core Count (12,288): Utilizing a standard 6-transistor bitcell configuration, the core area is calculated as:
  256 rows × 8 bits × 6 transistors/cell = 12,288 transistors
- Overhead (~ 5,300): The peripheral circuitry accounts for approximately 30% of the total area. This estimate includes:
  i) Row Decoder (8-to-256): Logic gates and wordline drivers required to address 256 individual rows.
  ii) Column Circuitry: Differential sense amplifiers, write drivers, and precharge circuits required for the 8 bitline pairs.
- Total Estimate: 12,288 (Core) + 5,300 (Overhead) ~ 17,600 transistors

2. ROM: Unified Memory (Single Block: 1024 rows × 8 bits)

- Core Count (8,192): ROM architecture utilizes a high-density 1-transistor bitcell:
  1024 rows × 8 bits × 1 transistors/cell = 8,192 transistors
- Overhead (~ 20,800): Despite the efficient core, the overhead is substantial due to the decoder scaling. Addressing 1,024 unique rows necessitates a complex 10-to-1024 decoder and, critically, 1,024 individual wordline drivers. This results in an overhead that is approximately 2.5x larger than the storage core itself.
- Total Estimate: 8,192 (Core) + 20,800 (Overhead) ~ 29,000 transistors

3. ROM: Split (Banked) Memory (4 Banks × 256 rows × 8 bits)

- Core Count (8,192): The total storage capacity is identical to the Unified Memory configuration, resulting in an equivalent core count.
- Overhead (~ 5,800): This architecture minimizes overhead by sharing peripheral resources. Instead of a single massive decoder, the design employs

the smaller 8-to-256 decoder (identical to the SRAM block) and broadcasts the wordlines to all four banks. Additional logic is limited to:

i) Bank Select Logic: Multiplexers to route the output from the active bank.

ii) Bank Enable Signals: Simple logic to power-gate inactive banks.

- Total Estimate: 8,192 (Core) + 5,800 (Overhead) ~ 14,000 transistors
- Physical Trade-off: While logic area is minimized, this configuration incurs a routing overhead. The physical layout requires distributing the address bus to four separate macros and routing four distinct 8-bit output buses to the central multiplexer, consuming additional metal layer area compared to the monolithic block.

4. Appendix – Activation Functions Mapping

Activation Function	Logical Role & Behavior	Critical Application	Risk of Exclusion
ReLU (Rectified Linear Unit)	Pass-through / Zeroing	Computer Vision: The standard for image recognition (e.g., smart security sensors). Computationally efficient for deep networks.	PE would be inefficient and slow for standard image tasks.
Identity	Pass-through	Residual Learning: Crucial for "Skip Connections" in deep networks (e.g., ResNet) to prevent vanishing gradients, and for Linear Output Layers in regression tasks.	The PE would fail to support modern deep architectures (ResNets) or regression models, severely limiting its flexibility and accuracy on complex tasks.
Tanh (Hyperbolic Tangent)	Zero-Centered Mapping	Voice Processing: Essential for the "Cell State" in recurrent networks used in voice activation (e.g., smart toys).	PE would struggle to maintain long-term memory, failing at sequence tasks.
Sigmoid	Probability Switch	Required for "Forget/Update Gates". Decides what information to keep.	PE could not run standard LSTMs or GRUs without Sigmoid, breaking voice/audio applications.
GELU (Gaussian Error Linear Unit)	Probabilistic Smoothing	Modern AI (Transformers)	PE would be stuck supporting only older (pre-2018) models
SiLU (Sigmoid Weighted Linear Unit)	Self-Gating, allows small negative values to propagate	The standard activation for MobileNetV3 and EfficientNet, which are specifically designed for low-power edge devices	PE could not natively accelerate the most popular "mobile-optimized" networks, forcing them to fallback to less accurate approximations.

Table 2: Activation Functions.